A Chained Machine Learning Approach to Motivate Retro-Cladding of Residential Buildings

Paul Nicholas[1], Yu Chen[2], Nihit Borpujari[3], Nitsan Bartov[4], Andreas Refsgaard[5]
[1,2,3,4]Royal Danish Academy, School of Architecture [5]Inflation
[1]paul.nicholas@kglakademi.dk [2,3,4]{yuch1949|nibo1955|niba2041}@edu.kglakademi.dk [5]mail@andreasrefsgaard.dk

This paper investigates how a novel approach to visualisation could help address the challenge of motivating residential retrofitting. Emerging retrofitting research and practice emphasises retro-cladding - the upgrading of the exterior facade of a building - using a modular approach. We present a machine-learning based approach aimed to motivate residential retrofitting through the generation of images and cost/benefit information describing climatically specific additions of external insulation and green roof panels to the façade of a Danish type house. Our approach chains a series of different models together, and implements a method for the controlled navigation of the principle generative styleGAN model. The approach is at a prototypical stage that implements a full workflow but does not include numerical evaluation of model predictions. Our paper details our processes and considerations for the generation of new datasets, the specification and chaining of models, and the linking of climatic data to travel through the latent space of a styleGAN model to visualise and provide a simple cost benefit report for retro-cladding specific to the local climates of five different Danish cities.

Keywords: *Retrofitting, Machine Learning, Generative Adversarial Networks, Synthetic Datasets*

INTRODUCTION

Hiding just behind the C40's ambition that all new buildings be net zero by 2030 is a second, larger ambition: that all buildings be net zero by 2050 [1]. Currently, energy use in buildings accounts for almost 40% of all CO2 emissions in the European Union. The long life of buildings means that the majority of today's buildings were built before the current era of low energy regulation. This means that while discussions of low energy architecture often have focus on new commercial buildings, the largest potential for energy reductions are actually found within existing non-commercial buildings. In this context, retrofitting of suburban housing is considered to be one of the most effective strategies for reducing energy consumption and increasing resilience to climate change. Over the next 10 years, it is estimated that retrofitting of existing buildings will ac-

count for 70% of total construction work in Denmark, with a majority of this work undertaken on suburban homes. This context is captured in the idea that 'the home of the future is the one that you already live in'.

In this paper we describe a visualisation tool aimed to motivate residential retrofitting. The tool focuses on retro-cladding - upgrading the exterior facade of a building through fitting of new elements - as façades and roofs play a central role in heat transfer and water management. . The tool takes the form of a dashboard that 1) visualises the adapted state of an existing residential building corresponding to average monthly weather, and 2) provides simplified 'what if' analytics that decrease uncertainty about the investment required and benefits achieved, specific to particular Danish cities.

We detail a novel modelling approach, which leverages the generalizability of ML and the possibility to chain together a series of different ML-based models. Our approach is chosen to address a base case for retrofitting in Denmark: the retrofitting of 1960s era residential developer-built housing, known as 'typhuse'. This case replicates the same house throughout Denmark and across different climatic zones. While a wide variety of expert architectural simulation tools and methods are already present in the design workflow for new-build, these tools are not directly transferable to the retrofitting context, where non-expert household level decision-making is a central driver of change. Here, research has identified owner engagement and economic viability are principal barriers, due to uncertainty about the investment required and the potential benefits achieved (Pardo-Bosch et al. 2019, El-Darwish & Gomaa, 2017).

To describe this modelling approach, our paper details our processes and considerations for the generation of new datasets, the specification and chaining of models, and the linking of climatic data to travel through the latent space of a styleGAN model to visualise and provide a simple cost benefit report for retro-cladding specific to the local climates of five different Danish cities.

A MACHINE-LEARNING BASED APPROACH

Our visualisation tool is developed using several different Machine Learning models, including Generative Adversarial Networks (GAN) and neural nets. These models present significant challenges to the architectural design workflow, where modelling, visualization and simulation tools are typically deployed to support one-off design.

Tamke et al. 2018 provides a broad overview of the emerging trajectories of application of Machine Learning for Architectural Design. In the field of image generation, styleGANs have been trained on architectural drawings related to satellite urban plan, building plan, and section as input training data to generate 2d drawings (Zhang 2019). cGANs have been trained on low resolution scans of hand drafted plan drawings, to generate new plan drawings (Chalíou 2017 [2], Kvochick 2018). Other contemporary research has explored the image to image translation to shortcut CFD simulation (Galanos et al 2019, Mokhtar et al 2020, Musil et al 2019), using a 2d ground floor plan input to represent building boundaries, and Nicholas et al 2020 describes the use of Pix2Pix to generate image-based predictions of material behaviour as a basis for design and fabrication. We are not aware of other approaches that use machine learning to visualise adaptations to existing buildings.

Generating synthetic datasets

The training of machine learning-based models requires a large data set of images. For architectural applications these cannot be scraped off the internet or accessed in a pre-existing database, but need to be generated from scratch. Similar to the approach taken in some computer vision tasks in which labels are difficult to generate, such as pose estimation (Tjaden 2017), synthetic architectural images can be created from CAD models. In conjunction with automation, this technique can be used to make large high quality datasets relatively quickly (Sun et al 2015). Render quality is an important consideration: Jabbar et al 2017 test Unity 3D (a realtime ren-

derer) and Blender (a photorealistic renderer which is slower), and show that for an image segmentation model, more photorealistic images lead to much improved performance. The self-generation of synthetic datasets has multiple advantages: Zhang 2019 identifies improved control over the features in the images, while Rossi and Nicholas 2020 demonstrate how an image representation can be used abstractly to encode multiple types of information across its different channels.

Generative Adversarial Networks

The basis of our workflow are two GAN models - a form of generative model commonly used to create new images. In GANs the generative model is pitted against an adversary: a discriminative model that learns to determine whether a sample is from the model distribution or the data distribution- (Goodfellow et al, 2014). The advantage of using GANs is that they are flexible and can be trained on anything that can be represented as an image. Because GANs learn a loss that adapts to the data, they can be applied to a multitude of tasks that traditionally would require explicit feature description and very different kinds of loss functions (Isola et al, 2018).This allows for simple and generic inputs to quickly yield a design space that goes beyond the training data.

Navigating Latent Space

During the training process, the model learns underlying patterns from the image dataset and simplifies them into a compressed but high dimensional representation known as the latent space. After the model is trained, points in the latent space are sampled to generate new images. The ability to control the output of a GAN model to achieve a desired image occurs through the sampling and navigation of the latent space. A significant limit is that controlled movement is difficult to achieve as the features in the latent space are normally very entangled. However since their appearance, it has been known that semantically meaningful vector operations can be made in a GAN's latent space. Modes of sampling the latent space can include uploading an image and finding a closest match, taking a random walk, or tracking along particular vectors. Radford et al 2015, for example, demonstrate projection and vector arithmetic in the latent space, as well as interpolation along a vector defined by two points in the latent space. A problem for recent research has been that finding meaningful movements through the latent space has required supervised approaches and human labelling. Recently, new unsupervised methods for finding semantically meaningful controls for movement in latent space have emerged, including Voynov & Babenko 2020 [3] and Härkönen et al 2020. In this research, we use the method developed by Härkönen et al 2020, which applies Principal Components Analysis (PCA) in the activation space of a styleGAN model to identify latent directions without any supervision.

Chaining Machine Learning Models

Our ML modelling pipeline is developed as a sequential model chain. In a chained model, single models encode specific independent operations as discrete modules with an input and output, and self-contained modules can be connected together. Chaining multiple models together sequentially enables a continuous modelling pipeline, with each module specific to a particular aspect of generation/prediction.

The advantage of this approach is that model architectures can be matched specifically to tasks, rather than trying to predict all relevant aspects via a single model. However with more models, the need for more training data is compounded. We circumvent this problem by training multiple models on the same base set of training data (see figure 1). Different combinations of image and labelling are used for the different models - this is further described in the case presented below.

This research moves beyond state of the art in a number of ways. Our model is developed for a specific existing building. Rather than aiming to generate a single image, our approach generates image sequences in which critical aspects of the content of

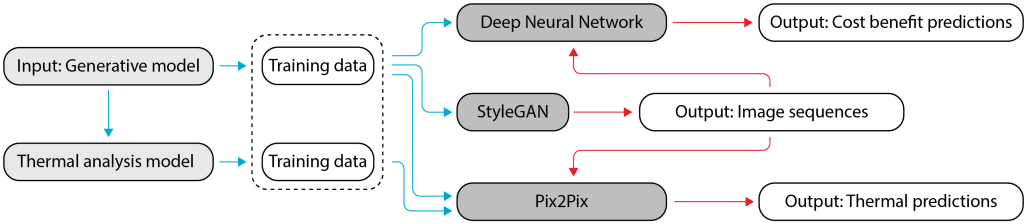


Figure 1
Flow diagram for chained models

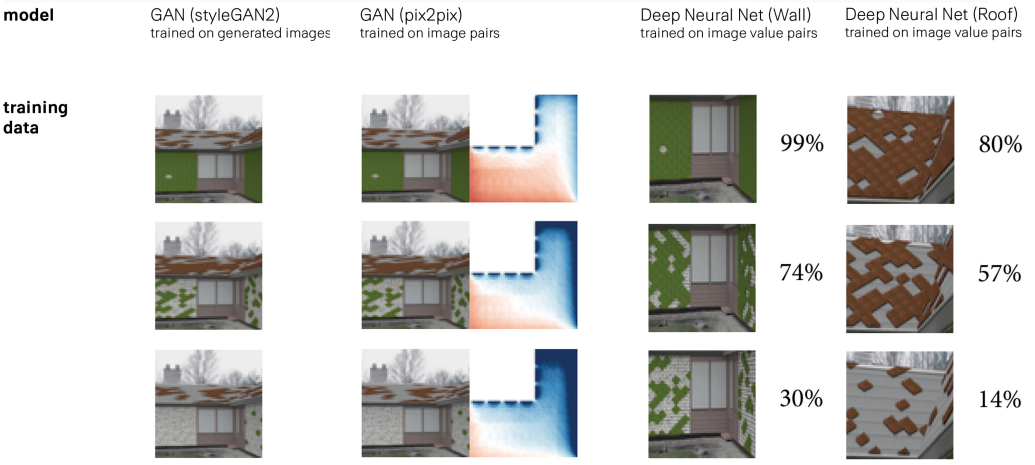


Figure 2
Chained ML models

each image are controlled. We implement a method for controlling movement through the latent space of a GAN model via architecturally relevant parameters, specifically climate related parameters. Model-chaining is used for both image generation and the evaluation of the generated images. Our evaluation approach allows us to search the GAN latent space to find specific series of images. The development and testing of this approach is still in progress.

EXPERIMENT

Our method is developed against a specific case: the Johan Christensen typehus (see figure 3), of which approximately 20,000 were built across Denmark from the mid 1960s.



Figure 3
The Johan Christensen typehus

To generate adapted versions of this house showing

different configurations of wall and roof cladding, we train three different kinds of machine learning-based model (see figure 2). A styleGAN model learns to generate new images of roof and wall panel configurations onto the base façade. A second type of GAN known as a Pix2Pix model learns to generate energy simulations that correspond to the facade and roof cladding configurations generated by the styleGAN. Lastly, two deep neural networks learn to predict numerical values including economic cost and yearly CO2 absorbed, taking the styleGAN-produced image as input.



Figure 4
Parametric modelling for wall and roof coverage iteration

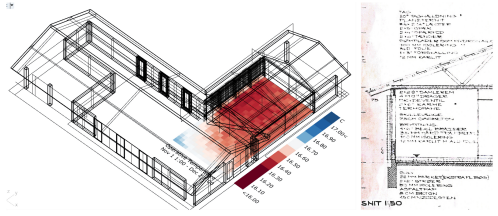


Figure 5
Modelling the baseline thermal context



Figure 6
StyleGAN generated images



Figure 7
Pairing rendered images with thermal simulations

To produce these images, we build a parametric model that generates different percentages of roof and wall coverage (see figure 4). It is important to note that this modelling and visualization approach does not aim to accurately reproduce the actual appearance of the façade and roof claddings. As claddings of differing thickness are visually identical, this variable is instead converted to visually communicate a panelisation of varying coverage. An initially full coverage of diamond panels is randomly reduced on the basis of two variables of "reduction" and "seed". The reduction variable controls the degree of reduction in panel coverage while the seed determines the reduction pattern. By randomly varying the input for the variables a dataset of 780 unique configurations is generated. Each time a configuration is made it triggers the rendering process within the definition and saves a file with the respective panel type and percentage of coverage. A dataset is made from these saved renders, as well as associated values stating the percentage of wall and roof coverage.

As the parametric model is automatically varied, an associated thermal model is also updated (see figure 5). This model is parameterized using geometric and material specifications taken from original drawings of the Johan Christensen house. With each iteration, the percentage of façade coverage is interpreted as a variable thickness material that completely covers the wall surface, excluding openings. In this conversion, 100% equates to 200mm thickness of insulation. In the context of building and construction, the thermal resistance per unit area of a material is defined by its R-value. The material definition within the thermal simulation allows for such specifications. We use an R value of 3 . These parameters are automatically input into the simulation. To produce a generalized simulation result, we simulate all versions with a base thermal context and measure change from a baseline of no added insulation. Simulations are performed on a simulation grid of size 64*64, and results are saved as an image paired with its respective render.



Figure 8
Moving along the principal direction for wall panels

The online machine learning platform Runway was used to train the styleGAN2 model, from a dataset of 780 render images. The styleGAN is able to generate new images that capture many of the visual qualities of the training images, but also introduce new qualities not in the original data - for example, tiles of differing sizes, differing distributions and differing densities (see figure 6). To train the Pix2Pix model, the renders generated for the styleGAN training set are paired with their associated thermal simulations (see figure 7). These images are downsampled to 64*64 and trained using a custom code running on Jupyter Notebooks. From this paired data, the pix2pix model learns to generate new heatmaps of the thermal environment from new rendered images of the building created by the styleGAN.

The final models to be trained are two Deep Neural Networks that predict a value when they are shown an image. Both networks are trained from the same render dataset used to train the styleGAN, but each extracts a different region of interest from this image (see figure 9). The network predicting roof coverage crops the image to the extents of the roof, and then resizes this crop to a 64*64 image. The network predicting wall coverage crops the image to the extents of the facade, and then resizes this crop to a 64*64 image. Both networks are trained using a custom code running on Jupyter Notebooks.

Finding directions in latent space

To control the transformation of specific features within image sequences, we use the GANspace approach developed by Harkonen et al 2020, which finds important latent directions based on Principal Component Analysis (PCA) applied in the latent space. Using this approach, we are able to identify the two distinct directions in the styleGAN model primarily related to wall and roof paneling. A series of four numbers are used to describe the settings for the wall and roof directions that determine the layers activated within the GANspace.

For example, for initiating a point to generate an image within the latent space we input a 'seed' value and define an edit. The edit specifies the principal direction, the first layer of activation, the last of activation, the movement along the principal direction interpreted as a -10 to 10 scale, and a direction. An initial edit is made to move to a set start point, and subsequent edits define incremental movement along the principal direction (see figure 8).



Figure 9
Extracting regions of interests for Deep Neural Networks

The same trajectories can be travelled from any seed point. The choice of a seed is considered a pivotal one that affects the search within the latent space drastically. An exploration of seeds as a departure point quickly reveals the entangled nature of the latent space as a majority of the seeds assessed draw the images away from their type house representation and into increasingly abstract collages of pixels and form, often influenced by the number and intensity of moves made (see figure 10). The challenges of navigation within the latent space is thus compounded by the search for a suitable seed.

Figure 10
Challenges of navigating latent space



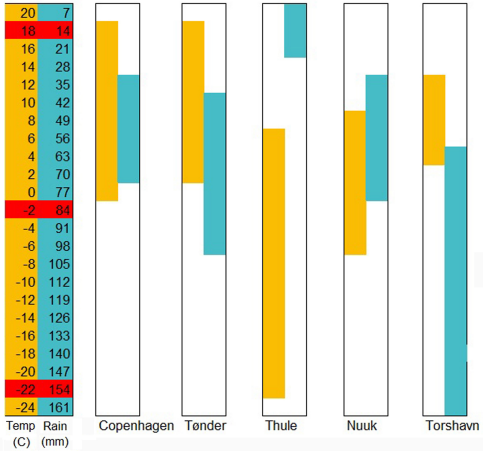
Figure 11
Cities range of yearly weather change

Finding yearly city climates in the latent space

Together with finding a suitable seed, navigation requires travel along the green roof and wall vectors. To relate climate parameters to travel in the latent space, we link progression along the principle directions to Danish city climate contexts. We choose two variables - monthly average rainfall and monthly average temperature - and associate them with the corresponding directions in the latent space that control the wall and roof panel coverage. We have developed this mapping using five city climate contexts that differ from one another in their relative change with regards to temperature and rain. The cities are Copenhagen, Tønder, Thule, Nuuk, and Tórshavn.

The maximum and minimum weather parameters between all cities are used as a bounds to map onto the full range of panel and roof coverage (0-100%) (see figure 11). The bounds for temperature are set by Copenhagen (18 C) and Thule (-24.6 C), and the bounds for rainfall are set by Tórshavn (162mm) and Thule(4mm).

This mapping is used to support location and monthly specific predictions within the visualisation tool, by using values for monthly average rainfall and monthly average temperature to navigate through the latent space.



Once the mapping is established, we use it to search for suitable seeds that, when used as a start-point for travel along the two principal directions for roof and wall, result in images that visualise the appropriate coverage for roof and wall throughout the range. As the rates of change differ throughout the latent space, we generate multiple images across the range to find more linear gradients. To make this search, an iterative method is employed that runs through thousands of seed candidates, and assesses every resulting image sequence.

Table 1
Mapping wall and roof coverage to latent space

Copenhagen	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Temperature (C)	0	0	2	7	12	16	18	17	14	9	5	3
Rain (mm)	49	39	32	38	40	47	71	66	62	59	48	49
Wall coverage %	40	40	36	24	13	4	0	2	9	20	29	33
Roof coverage %	23	15	10	15	16	22	40	36	33	31	22	23
Wall position	1	1	2	5	7	9	10	10	8	6	4	3
roof position	5	6	7	6	6	5	2	3	3	4	5	5

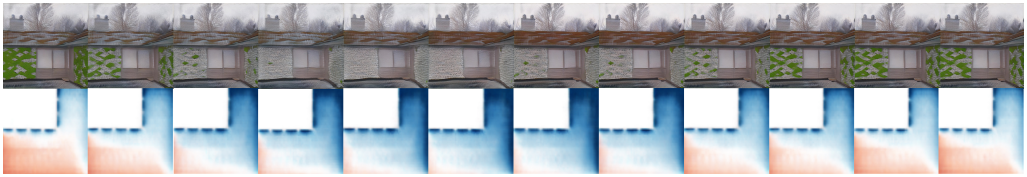


Figure 12
Image sequence for Copenhagen from January to December

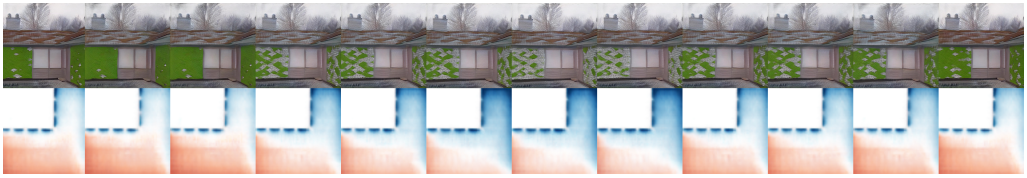


Figure 13
Image sequence for Nuuk from January to December

Once a seed is found to match the rate of change for a particular city within its range, the weather variables are mapped directly into the moves within the latent space to extract the precise images that depict the change in retrofit that reflect the average monthly weather (see table 1).

This sequence of images are then input into the pix2pix model to generate the respective thermal images (see figure 12 and 13), and the deep neural networks to enable estimated economic costs and environmental benefits. The deep neural networks predict the square meterage of green roof and façade paneling shown in each image. The estimate for green roof is multiplied by a given price per square metre to calculate overall cost, and by 2kg a square meter to estimate the quantity of CO2 that would be absorbed each year. The prediction of wall coverage is converted to a thickness of insulation and multiplied by a per meter cost. The benefit of this additional insulation is communicated visually by the thermal prediction. All calculations use current estimated prices within Denmark.

The entire modeling chain is then run in a simple web browser interface (see figure 14), allowing a non-expert user to choose cities and generate new images corresponding to specific months or periods of the year. New cities can be subsequently found in the model using the approach described above, and added to the web interface through specification of the seed and movement sequence.

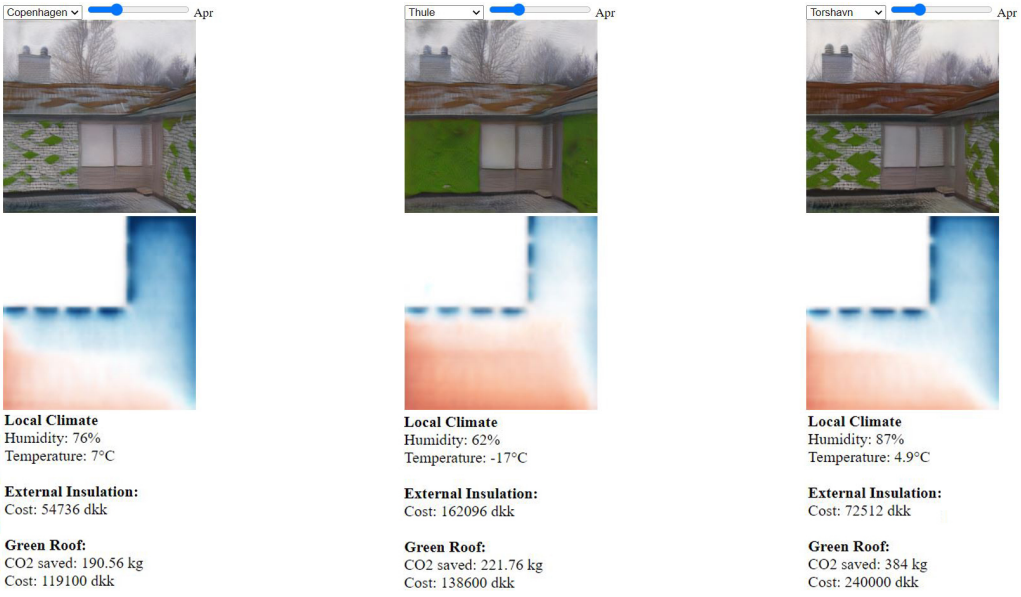
DISCUSSION AND CONCLUSION

This research is ongoing. To date, we have developed effective methods for generating large synthetic datasets of architectural images, and associations of renders, simulations and numeric information. We have also been successful in chaining together multiple machine learning models. This approach opens opportunities to connect , add or replace models for calculation, simulation, and representation within a prediction pipeline.

This has demonstrated that the approach of training multiple models, which will subsequently become inputs for one another, on the same data set, can be successful. Our case has particularly demonstrated the generalisability achieved by the Pix2Pix model, which is able to make good predictions from images generated by the styleGAN that are qualitatively different from those it has been trained on. The deep neural networks exhibited less generalisability when input with the same images. A next step is to gain a more precise evaluation of the predictive performance of each model.

The case we have tested is relatively simple, with

Figure 14
Browser interface



only two dimensions of change. Even so, we have had to choose representations that were not physically accurate but which allowed for learning. As we have self-generated the datasets, we have had full control over these features. The process of extracting these directions using the GANspace approach was also straightforward. However the ability to move along those directions in controllable ways was not straightforward, with the non-linearity of change being a particular and unforeseen problem. Here we have only begun to understand how to architecturally navigate the multiple dimensions and complexities of the trained models.

We believe that modeling workflows of this kind could have an important role to play in suburban retrofitting by leveraging the generalisability of ML, and the ability when working with ML models to frontload effort without needing all of the specifics.

For these types of architectural problem - multiple deployments exhibiting small amounts of variation - they offer a compelling alternative to an exhaustive modelling approach. In the specific context of our case, we don't need to know the cities in advance if we can find them later. Beyond our case, our method could extend as a means to model the adaptation of architectural typologies that are characterised by their mass deployment, versioning and repetition, for example urban squares, pocket parks, playgrounds, transport hubs, kiosks, as well as housing. Next steps in the project will further explore means for searching the GAN latent space for desired and controlled series of images, and performing numerical evaluation on our models.

REFERENCES

El-Darwish, I. and Gomaa, M. 2017, 'Retrofitting strategy for building envelopes to achieve energy efficiency. DOI: 10.1016/j.aej.2017.05.011', *Alexandria Engineering Journal*, 56, pp. 579-589

Galanos, T., Chronis, A., Vesely, O., Aichinger, A. and Koenig, R. 2019 'Best of both worlds – using computational design and deep learning for real-time urban performance evaluation', *1st International Con-*

ference on Optimization Driven Architectural Design
Goodfellow, I.J, Pouget-Abadie, J., Mirza, M., Xu, B., Warde-Farley, D., Ozair, S., Courville, D. and Bengio, y. 2014 'Generative adversarial nets', *NIPS*, p. Pages 2672–2680
Härkönen, E., Hertzmann, A., Lehtinen, J. and Paris, S. 2020 'Discovering Interpretable GAN Controls', *NeurIPS 2020*
Isola, P., Zhu, J., Zhou, T. and Efros, A.A 2017 'Image-to-Image Translation with Conditional Adversarial Networks', *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*
Jabbar, A., Farrawell, L., Fountain, J. and Chalup, S.K 2017 'Training Deep Neural Networks for Detecting Drinking Glasses Using Synthetic Images', *International Conference on Neural Information Processing*
Kvochick, T. 2018 'Sneaky Spatial Segmentation. Reading Architectural Drawings with Deep Neural Networks and Without Labeling Data', *Proceedings of the 38th Annual Conference of the Association for Computer Aided Design in Architecture (ACADIA)*
Mokhtar, S., Sojka, A. and Davila, C.C. 2020 'Conditional Generative Adversarial Networks for Pedestrian Wind Flow Approximation', *Proceedings of SimAUD 2020*
Musil, J., Knir, J., Vitsas, A. and Gallou, I. 2019 'Towards Sustainable Architecture: 3D Convolutional Neural Networks for Computational Fluid Dynamics Simulation and Reverse Design Workflow', *NeurIPS Workshop on Machine Learning for Creativity and Design 3.0*
Nicholas, P., Rossi, G., Williams, E., Bennett, M. and Schork, T. 2020, 'Integrating real-time multi-resolution scanning and machine learning for Conformal Robotic 3D Printing in Architecture', *International Journal of Architectural Computing*, 18, pp. 371-385
Pardo-Bosch, F., Cervera, C. and Ysa, T. 2019, 'Key aspects of building retrofitting: Strategizing sustainable cities', *Journal of Environmental Management*, 248, p. 109247
Peng, X. and Saenko, K. 2018 'Synthetic to Real Adaptation with Generative Correlation Alignment Networks', *IEEE Winter Conference on Applications of Computer Vision (WACV)*
Radford, A., Metz, L. and Chintala, S. 2016 'Unsupervised Representation Learning with Deep Convolutional Generative Adversarial Networks', *ICLR*
Rossi, G. and Nicholas, P. 2020 'Encoded Images: representational protocols for integrating cGANs in iterative computational design processes', *ACADIA 2020 : Distributed Proximities*
Sun, B., Peng, X. and Saenko, K. 2015 'Generating Large Scale Image Datasets from 3D CAD Models', *CVPR Workshop*
Tamke, M., Nicholas, P. and Zwierzycki, M. 2018 'Machine learning for architectural design: Practices and infrastructure', *International Journal of Architectural Computing*
Tjaden, H., Schwanecke, U. and Schömer, E. 2017 'Real-Time Monocular Pose Estimation of 3D Objects using Temporally Consistent Local Color Histograms', *In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*
Zhang, H. 2019, '3D Model Generation on Architectural Plan and Section Training through Machine Learning', *Technologies Special Issue "Computer-Aided Architectural Design"*, 7(4), p. 82
[1] https://www.c40.org/other/net-zero-carbon-buildings-declaration
[2] https://issuu.com/stanislaschaillou/docs/stanislas_chaillou_thesis_
[3] https://arxiv.org/abs/2002.03754